

Critical Values

A lab report is the most common artifact in healthcare and the least useful one. The engine that fixes that is built. The company around it is not.

Built today · 15 commits 11 MCP tools 121 tests Latency ~30 s / panel

Ground truth **unmeasured**

THESIS

The report is a dead end

Every blood test produces a PDF that the patient cannot read and the clinician has eight minutes to interpret. The information exists. The understanding does not.

Diagnostic volume is climbing on preventive and corporate screening, while lab chains have commoditised the test itself and now compete on price and turnaround. Nobody competes on *comprehension*, because until recently turning an unstructured report into structured clinical meaning was too expensive to do per patient.

That changed. Extraction, classification and grounded explanation now cost cents per report. The opportunity is not a chatbot that explains blood tests — that is already free. It is the intelligence layer that sits between the analyser and the human, and gets embedded in every channel where results are delivered.

POSITIONING

Infrastructure, not a destination app

The engine already does the hard part: severity classification against reference ranges, literature-grounded explanation, suggested next steps, and validated handling of the

literature grounded explanation, suggested next steps, and validated handling of the messy real-world cases — qualitative dipstick values, Turkish test names, missing bounds, units that disagree with the range.

The product decision is to sell that as a component to organisations that already own the patient relationship, rather than fight for consumer attention against a free substitute.

WHAT IT IS

An API and white-label layer that turns a raw lab report into structured, severity-routed, explained results — plus a clinician-facing brief. Sold per report to labs, telehealth and corporate health.

WHAT IT IS NOT

A consumer subscription, a doctor marketplace, or a diagnostic device. It surfaces information and reasoning; a clinician decides. That boundary is the regulatory position and it is load-bearing.

WHO PAYS

Four buyers, one wedge

Ranked by realistic time-to-first-rupee, not by market size. The first is the wedge because it is the only one where the buyer already holds both the data and the patient.

01

Diagnostic labs & chains

WEDGE

Lal PathLabs, Thyrocare, Metropolis, SRL. Thin differentiation, consumer-facing apps, and every report already flowing through their portal. ROI is engagement, retention and follow-up test upsell — measurable inside one quarter.

02

Telehealth platforms

STRONG

Eight-minute consults with scattered history. The pre-consult brief is directly valuable and sells as an API component, not a competing destination.

03

Corporate health & insurers

VIABLE

Bulk annual screening produces PDFs nobody opens. Buyer is an HR or actuarial budget rather than a clinician, which shortens the sale but weakens the clinical pull.

04

Hospitals

LATER

Real budgets, but 12–18 month cycles, security review, clinical validation and EMR integration. Requires references and capital you do not have yet.

SCOPE

What survives, what ships, what dies

The instinct with a working prototype is to add. Most of what follows is subtraction — three finished capabilities read better to a buyer than eight half-wired ones, and every cut below removes a dependency on somebody else's supply, liability or attention.

FEATURE DECISIONS

CAPABILITY	CALL	RATIONALE
MCP tool boundary	KEEP	A real process boundary makes each capability independently testable and swappable. It is also the integration surface a partner buys.
Severity routing & explainability payload	KEEP	Deviation, range, explanation, next step, sources. This is the product. Never ship a bare label.
Input & output validation layers	KEEP	Range sanity checks, dimensional unit parsing, rejection over guessing. This is what a clinical buyer's diligence actually probes.
Document ingestion — PDF and photo	ADD	Requiring a clean CSV means the product has no real input. Extraction from a lab PDF or a phone photo is the actual unlock.
Patient timeline & trajectory detection	ADD	A haemoglobin of 11.2 is a shrug; 14 → 12.5 → 11.2 is a workup. Single-point severity cannot see it, and nobody else is selling it.

Pre-consult clinician brief	ADD	Trends, abnormalities, open questions, cited to source documents. Valuable to both sides of the appointment and needs zero integrations to demo.
Accuracy evaluation harness	ADD	The dataset ships a <code>Status</code> column — ground truth, currently unused. Agreement measured at scale is the moat; the model is not.
PHI redaction & local-model path	ADD	Enterprise buyers block patient data leaving their estate. An on-premise inference option converts a hard no into a procurement conversation.
Cost controls — caching, tiered models	ADD	Unit economics kill more AI companies than accuracy. A lab at 100k reports/month needs cents, not the current several calls per panel.
Doctor directory	CUT	A supply-acquisition problem wearing an engineering costume. Marketplaces are won on supply, and it differentiates nothing.
Patient–doctor messaging	CUT	Commodity infrastructure that creates duty-of-care the moment a clinician replies. Revisit only as AI triage layered on someone else's inbox.
Autonomous appointment booking	REFRAME	Keep findings → specialty routing; drop the autonomy. Agent proposes, human confirms in one tap. Booking unsupervised is liability, not a feature.
Literature grounding on every result	NARROW	Keep for abnormalities, cache aggressively by test and direction. Paying a search plus two generations for a normal haemoglobin is indefensible at volume.
Consumer subscription	CUT	Testing is episodic, so retention is structurally poor, and the free substitute is one paste away. Reach patients through the lab that already has them.

BUILD ORDER

Five phases, each a standalone demo

Ordered by dependency, then by value per week. Phases one to three need nothing from the outside world, which is what protects the plan from stalling on somebody else's data or signature.

01

Persistence & identity

Everything downstream needs a patient with a history. This is the one layer that cannot be faked.

Postgres: patients, documents, results, decisions, audit_log

Move reference ranges into SQL with sex / age bands, source and updated_at

Load ranges into memory at MCP startup – DB is the authoring surface, not the hot path

Replace MemorySaver with a durable checkpointer so chat survives restarts

EXIT

A result submitted today is still there tomorrow, attributed to a patient, with every read and write audited.

02

Document ingestion

Accept what patients actually have: a lab PDF or a photo of a printout.

Multimodal extraction → LabRecord, reusing the existing schema and validators

Confidence per extracted field; anything low-confidence goes to human review, never silently through

Redaction pass before any outbound call; per-tenant provider policy

Ollama provider for a fully on-premise path

EXIT

Photograph a real report, get severity-routed explained results, with no data leaving the host when policy requires it.

03

Trends & the clinician brief

The differentiated core. Longitudinal reading is where clinical value concentrates, and the brief is how it gets delivered.

Trajectory detection: direction, rate of change, crossings into and out of range

New graph node between route and explain for panel-level correlation patterns

Generated brief: trends, abnormalities, medications, open questions – every claim cited to a source document

Severity magnitude score so results sort worst-first inside a band

EXIT

A clinician opens one page before an appointment and knows more

A clinician opens one page before an appointment and knows more than they would have learned in the first five minutes of it.

04

Measurement

The moat. Until this exists there is no answer to “why can't we build this ourselves?”

Score classification against the dataset's own Status column: confusion matrix, per-severity precision and recall

Publish the disagreement list – the interesting cases are the ones it gets wrong

LLM-as-judge rubric on a sampled set for explanation quality

Golden-set prompt regression, so a model swap cannot silently degrade output

Token and cost accounting per run, appended to the existing run log

EXIT

A single slide stating agreement, cost per report and p95 latency – with the method behind each.

05

Distribution

Only now does integration work pay off, because only now is there something worth integrating.

FHIR Observation and DiagnosticReport in and out

Tenant isolation, API keys, per-tenant range overrides and branding

Critical-value webhooks mirroring statutory panic-value notification

Clinician override capture – the feedback flywheel that compounds into the moat

EXIT

A design partner sends live traffic and receives results into their own portal without human intervention.

METRICS

Numbers you must be able to state

None of these exist yet. That is the honest current position, and establishing them is the substance of phase four – each one is a sentence a buyer or investor will ask for by name.

Agreement with ground truth

Per-severity precision and recall against clinician-labelled results. **The single most important number in the company.**

Cost per report

Fully loaded inference plus retrieval. **Must survive 100k reports/month in cents.**

Auto-clear rate

Share of results safely resolved without a human. Directly proportional to the value delivered.

Review time saved

Minutes per hundred results, measured against a control. **This is what the buyer is actually purchasing.**

p95 latency

Currently around 30s per panel and unoptimised. Batch tolerates it; interactive does not.

Override rate, trending down

How often a clinician disagrees. Falling over time is the flywheel working.

RISK

Graded on its own scale

Scored with the taxonomy the engine itself uses. Critical means it invalidates the plan; warning means it constrains it; normal means it is understood and priced in.

CRITICAL**No defensibility against the buyer building it themselves**

A lab's own team could assemble a competent version in a quarter. The model layer is commoditising and confers nothing.

MITIGATION Validated accuracy at scale, regulatory posture, integration depth, and the override flywheel. Phase four is not optional — it is the answer to this question.

CRITICAL**Liability for a missed critical value**

The first question every clinical buyer asks. An unanswerable version of it ends the sale.

MITIGATION Surface-and-explain framing, mandatory sign-off on criticals, escalation on low confidence, and a complete audit trail. Never auto-clear anything abnormal.

WARNING**Unit economics at volume**

Several LLM calls plus a literature search per panel does not survive contact with 100k reports a month.

MITIGATION Cache by test and direction, reserve the expensive path for abnormals, tier to smaller models for routine cases, batch overnight loads.

WARNING**Regulatory classification**

Software influencing care decisions attracts CDSCO attention in India, and FDA or MDR abroad. Suggesting next steps sits near the device boundary.

MITIGATION Stay on the information side of the line deliberately and in writing. Take advice before the first paid deployment, not after.

WARNING**Provider dependence**

One quota exhaustion took the primary model out today and forced a mid-session swap with no way to tell whether output quality moved.

MITIGATION The provider abstraction and fallback chain already exist. Add the golden-set regression so a swap is measurable rather than a leap of faith.

NORMAL**Extraction accuracy on real documents**